

Nikolay Nikolov

AI Scientist and Engineer

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Reinforcement Learning, Robotics, Computer Vision, LLMs

Education

- Oct 2014 **Imperial College London**, BEng & MEng Electronic and Information Engineering
- Sep 2018 First Class Honors 74.3/100%; GPA: 4.0/4.0
Courses: *Robotics* ▪ *Machine Learning* ▪ *Computer Vision* ▪ *Control* ▪ *Operating Systems* ▪ *Compilers* ▪ *OOP* ▪ *Mathematics* ▪ *Networks* ▪ *Databases* ▪ *Algorithms and Data Structures*
- Sep 2017 **ETH Zurich**, MEng Exchange Student
- Sep 2018 Courses: *DL* ▪ *ML* ▪ *Probabilistic AI* ▪ *Dynamic Programming and Optimal Control* ▪ *Computer Vision*

Experience and Research

- May 2024 **INSAIT**, PhD Student, Sofia, Bulgaria
 - Present *Robotics Foundation Models with non-robotics data such as 3D, human videos, world models*
Developed Do Anything Model (DAM) - state-of-the-art robotics foundation model with 3D understanding that significantly outperforms π_0 and π_0 -FAST, while using 10x less data
- Jul 2023 **Freelance AI Consulting**, Sofia, Bulgaria
- May 2024 Interior design rendering via diffusion models, code generation with LLMs, and others
- Dec 2018 **Wayve**, AI Applied Scientist, London, UK
- Dec 2022 *Developed new state-of-the-art autonomous driving AI methods and deployed in central London*
 - Part of the initial seed team of 20. Helped build the tech of the company to a series B unicorn
 - Developed the first end-to-end off-policy offline Reinforcement Learning method that can drive in complex urban real-world environments such as London
 - Engineered and developed foundational models for Imitation Learning
 - Contributions to research such as *learning from human feedback*, *robot perception*, *multitask learning*, *temporal modelling*, *causal confusion*, *data manipulation*
 - Engineered core AI systems for training, monitoring and deployment
- Sep 2017 **Learning & Adaptive Systems Group**, Research Assistant, ETH Zurich
- Sep 2018 *Information-directed exploration in Reinforcement Learning*
 - Developed a stochastic Information-Directed RL method that significantly outperforms state-of-the-art
 - Supervisor: Prof. Andreas Krause; Paper published at ICRL 2019
- Jul 2017 **Ocado Technology**, Robotics Research Intern, Hatfield, UK
- Sep 2017 *Deep Learning for robot picking*
Implemented a deep RL system for picking objects from a basket and deployed on a UR10 robotic arm
- Jan 2017 **Dyson Robotics Lab**, Research Assistant, Imperial College London
- Sep 2017 *Bayesian Fusion for Volumetric SLAM based on Occupancy Mapping*
 - Developed and implemented volumetric Bayesian 3D reconstruction method from depth camera
 - Supervisor: Prof. Stefan Leutenegger. Paper published at ICRA 2018
- Jul 2016 **ARM**, Hardware Engineering Intern, Sheffield, UK
- Sep 2016 *Hardware verification for memory management units*

Publications

- PDF **ReVLA: Reverting Visual Domain Limitation of Robotic Foundation Models** *Sombit Dey*, Jan-Nico Zaeche*, Nikolay Nikolov*, Luc Van Gool*, Danda Pani Paudel**.
IEEE International Conference on Robotics and Automation (ICRA), 2025
- PDF **Urban Driving with Conditional Imitation Learning.** *J Hawke*, R Shen*, C Gurau*, S Sharma*, D Reda*, N Nikolov*, P Mazur*, S Micklethwaite*, N Griffiths*, A Shah*, A Kendall**.
IEEE International Conference on Robotics and Automation (ICRA), 2020
- PDF **Information-Directed Exploration for Deep Reinforcement Learning.**
Nikolay Nikolov, Johannes Kirschner, Felix Berkenkamp, Andreas Krause.
International Conference on Learning Representations (ICLR), 2019
- PDF **Efficient Octree-Based Volumetric SLAM Supporting Signed-Distance and Occupancy Mapping.** *Emanuele Vespa, Nikolay Nikolov, Marius Grimm, Luigi Nardi, Paul H J Kelly, Stefan Leutenegger.* IEEE International Conference on Robotics and Automation (ICRA), 2018

Skills

- Programming *Python ▪ C++ ▪ C ▪ SQL ▪ JavaScript ▪ Shell ▪ Java*
OOP, concurrent programming, algorithms, data structures, vectorized computation
- AI Reinforcement Learning, Computer Vision, Generative AI, LLMs, Imitation Learning, Supervised Learning, Unsupervised Learning
- ML Modelling Transformers, Diffusion, VLMs, VLAMs, CNNs, GANs, VAEs, Adversarial training, Ensemble models, Uncertainty models
- Deep Learning implementation, training, debugging, evaluation, fine-tuning, reproducing literature
- MLOps cloud training, cloud deployment, distributed training, training speed optimization
- Data Big Data, cleaning, analysis, collection, filtering, noisy data, biased datasets
- Software PyTorch ▪ transformers ▪ diffusers ▪ PySpark ▪ pandas ▪ numpy ▪ GCP ▪ Azure ▪ WandB
▪ PyArrow ▪ ROS ▪ OpenCV ▪ pybind ▪ CI/CD ▪ git ▪ docker ▪ Linux ▪ bazel ▪ ...

Selected Projects

- 2023 3D architectural model rendering with Generative AI
- 2022 Breaking causal confusion in data in autonomous driving
- 2022 Data manipulation for learning diverse skills in autonomous driving
- 2021 Offline Reinforcement Learning for autonomous driving in central London
- 2020 Learning from human feedback in autonomous driving
- 2019 Imitation Learning for autonomous driving in central London
- 2018 Open-source Deep Reinforcement Learning Library
- 2017 Deep Reinforcement Learning for robot picking
- 2017 Autonomous indoor robot mapping, localization and navigation

Miscellaneous

- Languages English ▪ Bulgarian ▪ Russian
- Citizenship Bulgarian (EU) ▪ British
- 2007-Present Aikido - 1st Dan Black Belt
- 2013 STEM distinction by the President of Bulgaria *Sofia, Bulgaria*
- 2013 International Young Physicists Tournament - Bronze Medal *Taipei, Taiwan*